

HEAT CAPACITY AND VISCOSITY OF GLASSES AND MELTS IN THE SYSTEM $\text{Al}_2\text{O}_3\text{-CaO-MgO-SiO}_2\text{-TiO}_2$

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Experimental measurements of heat capacity and modelling of heat capacity and viscosity of melts and glasses in the $\text{Al}_2\text{O}_3\text{-CaO-MgO-SiO}_2\text{-TiO}_2$ system have been presented for selected unary, binary, ternary and quaternary subsystems. Adiabatic and differential scanning calorimetry have been used to measure heat capacity of synthesized glasses at low, intermediate and high temperatures. A unified model for both heat capacity and viscosity of melts and glasses in the $\text{Al}_2\text{O}_3\text{-CaO-MgO-SiO}_2\text{-TiO}_2$ system has been developed on the basis of the configurational entropy theory [1], extensive experimental database for viscosity [2] and collected from literature and measured heat capacity data. First, heat capacity of glasses has been modelled at temperatures above 300 K using a simple polynomial expression. Second, viscosity of melts and glasses has been described over a wide temperature range via the configurational heat capacity and Adam-Gibbs theory. Compositional dependences of the heat capacity and viscosity parameters have been described by both linear and quadratic rules of mixing depending on availability and/or agreement with experimental data. Using the developed model both heat capacity and viscosity of melts and glasses can now be predicted over the wide temperature and compositional ranges in the $\text{Al}_2\text{O}_3\text{-CaO-MgO-SiO}_2\text{-TiO}_2$ system including supercooled melts and glasses.

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2. Kondratiev, A. OxiVis database of experimental viscosities in the oxide melts and glasses // 2004-2026. Available on request via al.v.kondratiev@gmail.com

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